```
=== Results ===
Method   Window   Pseudo         Q3      SOV
---------------------------------------------
gor1     21       5            62.50    57.90
gor1     17       0            62.40    58.00
gor1     17       1            62.40    58.00
gor1     17       5            62.40    57.90
gor1     17       10           62.40    57.90
gor1     21       0            62.40    57.90
gor1     21       1            62.40    57.90
gor1     21       10           62.40    57.80
gor1     11       0            61.70    58.40
gor1     11       1            61.70    58.50
gor1     11       5            61.70    58.40
gor1     11       10           61.70    58.40
gor1     7        0            60.70    57.10
gor1     7        1            60.70    57.10
gor1     7        10           60.70    57.40
gor1     7        5            60.60    57.30
gor3     11       10           59.00    48.00
gor3     17       10           59.00    45.80
gor3     11       5            58.40    46.80
gor3     17       5            58.40    44.40
gor3     7        10           58.30    50.80
gor3     11       1            58.00    45.60
gor3     17       1            58.00    43.80
gor3     7        5            57.90    49.50
gor3     11       0            57.90    45.30
gor3     21       10           57.90    44.80
gor3     17       0            57.70    43.40
gor3     21       5            57.50    43.40
gor3     7        1            57.30    47.50
gor3     21       0            57.20    42.80
gor3     21       1            57.20    43.00
gor3     7        0            57.00    46.80

Best: method=gor1 window=21 pseudocount=5 Q3=62.50 SOV=57.90
```

GOR I outperforms GOR III across the board (62.5% vs 59.0%), because GOR III suffers from more severe data sparsity — insufficient sample size in the central amino acid sub-matrices.

Window size: GOR I performs best at 17–21 and worst at 7. This indicates that larger windows capture more contextual information, though the difference between 21 and 17 is negligible.

Pseudocount: Has almost no effect on GOR I (sufficient data volume), but provides a slight improvement for GOR III, because its count matrices are sparser — pseudocounts help alleviate the zero-probability problem.

Optimal parameters: `gor1 window=21 pseudocount=5`, Q3 = 62.50

01 — Window Size Effect
Avg SOV by window size (all pseudocounts)
60.0
55.0
50.0
45.0
40.0
17
GOR I: 57.95
GOR III: 44.35
7
11
17
21
Window
GOR I
GOR III
02 — Pseudocount Effect
Avg SOV by pseudocount (all windows)
60.0
56.0
52.0
48.0
44.0
0
1
5
10
Pseudocount
GOR I
GOR III
03 — Q3 vs SOV Scatter
All 32 configs — two metrics correlated
64
62
60
58
56
Q3
42
47
52
60
SOV
GOR I
GOR III
Best
04 — Best Q3 per Window
Max Q3 achieved at each window size
64
61
58
55
62.
w=7
w=11
w=17
w=21
GOR I
GOR III
GOR I — pairwise information model
GOR III — single residue model
Best configuration: w=21, pseudo=5

Q3 Score
SOV Score
toggle metric for plots 1 & 2
01 — Window Size Effect
Avg Q3 by window size (all pseudocounts)
64.0
62.0
60.0
58.0
56.0
7
11
17
21
Window
GOR I
GOR III
02 — Pseudocount Effect
Avg Q3 by pseudocount (all windows)
64.0
62.0
60.0
58.0
56.0
0
1
5
10
Pseudocount
GOR I
GOR III
03 — Q3 vs SOV Scatter
All 32 configs — two metrics correlated
64
62
60
58
56
Q3
42
47
52
60
SOV
GOR I
GOR III
Best
04 — Best Q3 per Window
Max Q3 achieved at each window size
64
61
58
55
62.
w=7
w=11
w=17
w=21
GOR I
GOR III
GOR I — pairwise information model
GOR III — single residue model
Best configuration: w=21, pseudo=5